

第五次作业

1. (二维信号的傅里叶级数)

课程中提到, 二维平面的热传导方程如下:

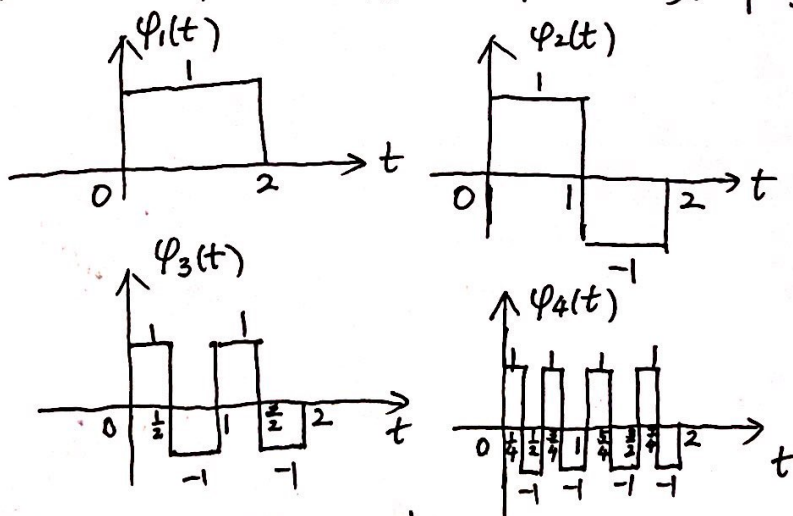
$$\begin{cases} \frac{\partial f(x, y, t)}{\partial t} = k \left[\frac{\partial^2 f(x, y, t)}{\partial x^2} + \frac{\partial^2 f(x, y, t)}{\partial y^2} \right] \\ f(x, y, 0) = f(x, y) \end{cases}$$

① 请用解一维~~傅里叶~~热传导方程的思路, 解以上二维热传导方程。

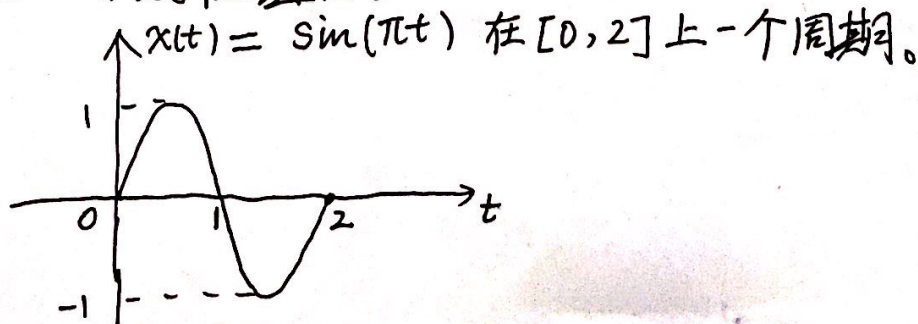
② 由此请推出二维信号傅里叶级数的表达式。

2. (哈尔小波分解信号)

哈尔小波是一组正交基, 它的前4个正交基函数 $\varphi_1(t)$, $\varphi_2(t)$, $\varphi_3(t)$, $\varphi_4(t)$ 如下:



① 请利用上述正交基, 将以下信号 $x(t)$ 分解为这4个正交基的线性组合。



即 $x(t) \approx a_1 \varphi_1(t) + a_2 \varphi_2(t) + a_3 \varphi_3(t) + a_4 \varphi_4(t)$
求 a_1, a_2, a_3, a_4 。

② (Matlab编程): 请在一张图上用不同颜色画出
 $x(t)$ 和 $a_1 \varphi_1(t) + a_2 \varphi_2(t) + a_3 \varphi_3(t) + a_4 \varphi_4(t)$, 看一下多大区别。
(提示: 由于计算机只能处理离散, 所以连续信号要转化为离散处理)

$t = [0: 0.001: 2];$ (0到2每隔0.001取一个点)

$y = x(t);$ (你可以写一个 Function $y = x(t)$)

$\text{plot}(x, y, 'k-');$ (可用 help 查看 plot 函数介绍)

③ (选做), 你能以此类推, 画出 $\varphi_5(t), \varphi_6(t) \dots \varphi_N(t)$ 吗?

① 若 $x(t) \approx \sum_{n=1}^N a_n \varphi_n(t)$, 请给出 a_n 的一般表达式。
($x(t) = \sin(\pi t)$ 时)

② 证明随着 N 增大, $\sum_{n=1}^N a_n \varphi_n(t)$ 对 $x(t) = \sin(\pi t)$ 在 $[0, 2]$ 的模拟越来越精确。即:

$$\lim_{N \rightarrow +\infty} \int_0^2 \left[x(t) - \sum_{n=1}^N a_n \varphi_n(t) \right]^2 dt = 0$$

3. (施密特正交化)

假设一组基 $\{a_1, a_2, \dots, a_N\}$ 不是正交基, 则可以对它们进行施密特正交化变换为一组正交基, 过程如下:

$$\beta_1 = a_1$$

$$\beta_2 = a_2 - \frac{\langle a_2, \beta_1 \rangle}{\langle \beta_1, \beta_1 \rangle} \beta_1$$



$$\beta_3 = a_3 - \frac{\langle a_3, \beta_1 \rangle}{\langle \beta_1, \beta_1 \rangle} \beta_1 - \frac{\langle a_3, \beta_2 \rangle}{\langle \beta_2, \beta_2 \rangle} \beta_2$$

⋮

$$\beta_N = a_N - \sum_{i=1}^{N-1} \frac{\langle a_N, \beta_i \rangle}{\langle \beta_i, \beta_i \rangle} \beta_i$$

请用数学归纳法证明, $\{\beta_1, \beta_2, \dots, \beta_N\}$ 是一组正交基,
即 $\langle \beta_i, \beta_j \rangle = 0 \quad (i \neq j)$

(提示: 需要用到内积 $\langle \cdot, \cdot \rangle$ 的4个性质)

4. (用施密特正交化将非正交基化为正交基)

① 已知 $a_1(t)=1$, $a_2(t)=t$, $a_3(t)=t^2$, $a_4(t)=t^3$ 在 $[0, 1]$ 上依照内积运算 $\langle x(t), y(t) \rangle = \int_0^1 x(t) \overline{y(t)} dt$ 是非正交基, 请用施密特正交化将 $\{a_i\}_{i=1 \sim 4}$ 化为正交基 $\{\beta_i\}_{i=1 \sim 4}$

② 请基于①中求出的正交基 $\beta_1(t), \beta_2(t), \beta_3(t), \beta_4(t)$ 将 $x(t)=t^5$ 进行正交分解, 即

$$x(t) \approx b_1 \beta_1(t) + b_2 \beta_2(t) + b_3 \beta_3(t) + b_4 \beta_4(t)$$

求 b_1, b_2, b_3, b_4

③ (Matlab编程): 请在一张图上用不同颜色画出 $x(t)$ 和 $b_1 \beta_1(t) + b_2 \beta_2(t) + b_3 \beta_3(t) + b_4 \beta_4(t)$, 看一下区别。



5. (有限长离散信号的傅里叶级数) (书上4.1节)

假设有一个有限长序列 $x[n]$ 只在 $n=0, 1, 2, \dots, (N-1)$ 这 N 个点上不为 0, 我们构造一串序列 $\{a_k\}_{k=0, 1, 2, \dots, (N-1)}$ 使得:

$$x[n] = \sum_{k=0}^{N-1} a_k e^{j\frac{2\pi}{N}nk} \quad (1)$$

请证明:

$$a_k = \frac{1}{N} \sum_{n=0}^{N-1} x[n] e^{-j\frac{2\pi}{N}nk} \quad (2)$$

提示: 请你首先证明 $\left\{ e^{j\frac{2\pi}{N}nk} \right\}_{k=0, 1, 2, \dots, (N-1)}$

~~是~~ 在 $\langle x[n], y[n] \rangle = \sum_{n=0}^{N-1} x[n] \overline{y[n]}$ 下是一组正交基, 即

$$\langle e^{j\frac{2\pi}{N}nk}, e^{j\frac{2\pi}{N}nl} \rangle = \begin{cases} 0, & \text{当 } k \neq l \text{ 时} \\ N, & \text{当 } k = l \text{ 时} \end{cases}$$

然后将 (1) 代入 (2) 即得。

6. (傅里叶变换的非复数形式)

对非周期函数 $x(t)$, 构造傅里叶变换的非复数形式如下:

$$X_{\cos}(\omega) = \int_{-\infty}^{+\infty} x(t) \cos(\omega t) dt$$

$$X_{\sin}(\omega) = \int_{-\infty}^{+\infty} x(t) \sin(\omega t) dt$$

① 请证明:

$$x(t) = \frac{1}{2\pi} \int_{-\infty}^{+\infty} [X_{\cos}(\omega) \cos(\omega t) + X_{\sin}(\omega) \sin(\omega t)] d\omega$$



② 书上的傅里叶变换复数形式为:

$$X(j\omega) = \int_{-\infty}^{+\infty} x(t) e^{-j\omega t} dt$$

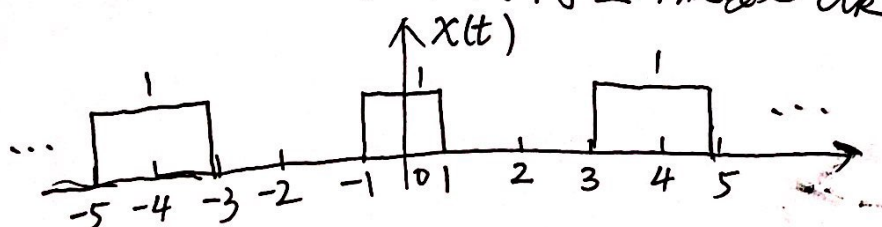
$$x(t) = \frac{1}{2\pi} \int_{-\infty}^{+\infty} X(j\omega) e^{j\omega t} d\omega$$

请用 $X_{\cos}(\omega)$ 和 $X_{\sin}(\omega)$ 表示 $X(j\omega)$, 也请用 $X(j\omega)$ 表示 $X_{\cos}(\omega)$ 和 $X_{\sin}(\omega)$ 。

③ (只想不做): ①和②说明, $X_{\cos}(\omega)$ 和 $X_{\sin}(\omega)$ 表达的思想内涵与 $X(j\omega)$ 完全相同。请思考, 如果用 $X_{\cos}(\omega)$ 和 $X_{\sin}(\omega)$ 来代替 $X(j\omega)$, 则第三章3.4节中傅里叶变换的性质将会怎样被改写? 从中体会傅里叶变换复数形式的方便。

7. (Matlab 编程 吉布斯现象)

① 求下列周期方波的傅里叶级数 a_k



② ~~请证明~~ 基于① 请证明:

$$x(t) = \sum_{k=-\infty}^{+\infty} a_k e^{jk\omega_0 t}$$

$$(\omega_0 = \frac{2\pi}{T_0} = \frac{2\pi}{4} = \frac{\pi}{2})$$

$$= \sum_{k=-\infty}^{+\infty} \frac{\sin(k\frac{\pi}{2})}{k\pi} e^{jk\frac{\pi}{2}t}$$

$$= \frac{1}{2} + \sum_{k=1}^{+\infty} \frac{2\sin(k\frac{\pi}{2})}{k\pi} \cos(k\frac{\pi}{2}t)$$



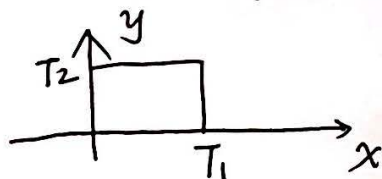
我们假设 $x_N(t) = \frac{1}{2} + \sum_{k=1}^N \frac{2 \sin(k\frac{\pi}{2})}{k\pi} \cos(k\frac{\pi}{2}t)$

请用 Matlab 画出 $N=5, 10, 100, 1000$ 时的 $x_N(t)$ 图, 并与 $x(t)$ 的图对比, 观察吉布斯现象。



第五次作业答案

1. 设 $f(x, y)$ 是长为 T_1 , 宽为 T_2 的平板



$$w_1 = \frac{2\pi}{T_1} \quad w_2 = \frac{2\pi}{T_2}$$

$$\begin{aligned} \text{若 } f(x, y) = & \sum_{k_1=0}^{+\infty} \sum_{k_2=0}^{+\infty} A_{k_1 k_2} \cos(k_1 w_1 x) \cos(k_2 w_2 y) + \\ & \sum_{k_1=0}^{+\infty} \sum_{k_2=1}^{+\infty} B_{k_1 k_2} \cos(k_1 w_1 x) \sin(k_2 w_2 y) + \\ & \sum_{k_1=1}^{+\infty} \sum_{k_2=0}^{+\infty} C_{k_1 k_2} \sin(k_1 w_1 x) \cos(k_2 w_2 y) + \\ & \sum_{k_1=1}^{+\infty} \sum_{k_2=1}^{+\infty} D_{k_1 k_2} \sin(k_1 w_1 x) \sin(k_2 w_2 y) \end{aligned}$$

则

$$f(x, y, t) = f(x, y) e^{-K(k_1^2 w_1^2 + k_2^2 w_2^2) t}$$

运用一维傅里叶级数推导方法可得:

$$\left\{ \begin{aligned} A_{00} &= \frac{1}{T_1 T_2} \int_0^{T_1} \int_0^{T_2} f(x, y) dx dy \\ A_{0k_2} &= \frac{2}{T_1 T_2} \int_0^{T_1} dx \int_0^{T_2} f(x, y) \cos(k_2 w_2 y) dy \\ A_{k_1 0} &= \frac{2}{T_1 T_2} \int_0^{T_2} dy \int_0^{T_1} f(x, y) \cos(k_1 w_1 x) dx \\ A_{k_1 k_2} &= \frac{4}{T_1 T_2} \int_0^{T_1} \int_0^{T_2} f(x, y) \cos(k_1 w_1 x) \cos(k_2 w_2 y) dx dy \\ &\quad (k_1 \neq 0 \text{ 且 } k_2 \neq 0) \end{aligned} \right.$$



$$\begin{cases} B_{0k_2} = \frac{2}{T_1 T_2} \int_0^{T_1} dx \int_0^{T_2} f(x, y) \sin(k_2 w_2 y) dy \\ B_{k_1 k_2} = \frac{4}{T_1 T_2} \int_0^{T_1} dx \int_0^{T_2} f(x, y) \cos(k_1 w_1 x) \sin(k_2 w_2 y) dy \\ (k_1 \neq 0) \end{cases}$$

$$\begin{cases} C_{k_1 0} = \frac{2}{T_1 T_2} \int_0^{T_2} dy \int_0^{T_1} f(x, y) \sin(k_1 w_1 x) dx \\ C_{k_1 k_2} = \frac{4}{T_1 T_2} \int_0^{T_1} dx \int_0^{T_2} f(x, y) \sin(k_1 w_1 x) \cos(k_2 w_2 y) dy \\ (k_2 \neq 0) \end{cases}$$

$$D_{k_1 k_2} = \frac{4}{T_1 T_2} \int_0^{T_1} dx \int_0^{T_2} f(x, y) \sin(k_1 w_1 x) \sin(k_2 w_2 y) dy \\ (k_1 \neq 0 \text{ 且 } k_2 \neq 0)$$

也可用复数表示法

$$a_{k_1 k_2} = \frac{1}{T_1 T_2} \int_0^{T_1} dx \int_0^{T_2} f(x, y) e^{-j(k_1 w_1 x + k_2 w_2 y)} dy$$

$$f(x, y) = \sum_{k_1=-\infty}^{+\infty} \sum_{k_2=-\infty}^{+\infty} a_{k_1 k_2} e^{j(k_1 w_1 x + k_2 w_2 y)}$$



$$2. \textcircled{1} \quad \langle \varphi_i, \varphi_j \rangle = \int_0^2 \varphi_i(t) \overline{\varphi_j(t)} dt$$

$$= \begin{cases} 2 & i=j \\ 0 & i \neq j \end{cases}$$

$$a_i = \frac{\langle x, \varphi_i \rangle}{\langle \varphi_i, \varphi_i \rangle} = \frac{1}{2} \langle x, \varphi_i \rangle$$

$$a_1 = \frac{1}{2} \langle x, \varphi_1 \rangle = \frac{1}{2} \int_0^2 \sin(\pi t) dt = 0$$

$$a_2 = \frac{1}{2} \langle x, \varphi_2 \rangle = \frac{1}{2} \left[\int_0^1 \sin(\pi t) dt - \int_1^2 \sin(\pi t) dt \right]$$

$$= \frac{2}{\pi}$$

$$a_3 = \frac{1}{2} \langle x, \varphi_3 \rangle = \frac{1}{2} \left[\int_0^{\frac{1}{2}} \sin(\pi t) dt - \int_{\frac{1}{2}}^1 \sin(\pi t) dt + \right.$$

$$\left. \int_{\frac{3}{2}}^2 \sin(\pi t) dt - \int_{\frac{3}{2}}^{\frac{5}{2}} \sin(\pi t) dt \right]$$

$$= 0$$

$$a_4 = \frac{1}{2} \langle x, \varphi_4 \rangle = \frac{1}{2} \left[\int_0^{\frac{1}{4}} \sin(\pi t) dt - \int_{\frac{1}{4}}^{\frac{1}{2}} \sin(\pi t) dt + \right.$$

$$\int_{\frac{3}{4}}^{\frac{5}{4}} \sin(\pi t) dt - \int_{\frac{5}{4}}^{\frac{3}{2}} \sin(\pi t) dt +$$

$$\int_{\frac{7}{4}}^{\frac{9}{4}} \sin(\pi t) dt - \int_{\frac{9}{4}}^{\frac{7}{2}} \sin(\pi t) dt \left. \right]$$

$$= 0$$

② 见后续程序。

③ (a) 除了 $a_2 = \frac{2}{\pi}$ 之外, 其它 a_i 都为 0



3. 证明: ① $N=1$ 时, $\beta_1 = \alpha_1$, 是正交基, 命题成立

② 若 $N=k$ 时成立, 即 $\langle \beta_i, \beta_j \rangle = 0$, 其中 $i \neq j$ 且 $i, j \leq k$ 时,

要证 $N=k+1$ 时成立。

只须证 $\langle \beta_{k+1}, \beta_j \rangle = 0$ 其中 $j=1 \sim k$

证明: 由于

$$\beta_{k+1} = \alpha_{k+1} - \sum_{i=1}^k \frac{\langle \alpha_{k+1}, \beta_i \rangle}{\langle \beta_i, \beta_i \rangle} \beta_i$$

$$\text{则 } \langle \beta_{k+1}, \beta_j \rangle \quad (j=1 \sim k)$$

$$= \langle \alpha_{k+1}, \beta_j \rangle - \sum_{i=1}^k \frac{\langle \alpha_{k+1}, \beta_i \rangle}{\langle \beta_i, \beta_i \rangle} \langle \beta_i, \beta_j \rangle$$

(这一步用到内积的齐次性和叠加性)

$$= \langle \alpha_{k+1}, \beta_j \rangle - \frac{\langle \alpha_{k+1}, \beta_j \rangle}{\langle \beta_j, \beta_j \rangle} \langle \beta_j, \beta_j \rangle$$

(这一步用到 $\langle \beta_i, \beta_j \rangle = 0$ 当 $i \neq j$ 时)

$$= \langle \alpha_{k+1}, \beta_j \rangle - \langle \alpha_{k+1}, \beta_j \rangle = 0$$

命题成立。



$$4. \textcircled{1} \quad \beta_1 = a_1 = 1$$

$$\beta_2 = a_2 - \frac{\langle a_2, \beta_1 \rangle}{\langle \beta_1, \beta_1 \rangle} \beta_1$$

$$= t - \frac{\langle t, 1 \rangle}{\langle 1, 1 \rangle} \cdot 1$$

$$= t - \frac{\int_0^1 t dt}{\int_0^1 1 dt} \cdot 1$$

$$= t - \frac{1}{2}$$

$$\beta_3 = a_3 - \frac{\langle a_3, \beta_1 \rangle}{\langle \beta_1, \beta_1 \rangle} \beta_1 - \frac{\langle a_3, \beta_2 \rangle}{\langle \beta_2, \beta_2 \rangle} \beta_2$$

$$= t^2 - \frac{\langle t^2, 1 \rangle}{\langle 1, 1 \rangle} \cdot 1 - \frac{\langle t^2, t - \frac{1}{2} \rangle}{\langle t - \frac{1}{2}, t - \frac{1}{2} \rangle} (t - \frac{1}{2})$$

$$= t^2 - \frac{1}{3} - \frac{\frac{1}{4} - \frac{1}{3} \times \frac{1}{2}}{\frac{1}{12}} (t - \frac{1}{2}) = t^2 - t + \frac{1}{6}$$

$$\beta_4 = a_4 - \frac{\langle a_4, \beta_1 \rangle}{\langle \beta_1, \beta_1 \rangle} \beta_1 - \frac{\langle a_4, \beta_2 \rangle}{\langle \beta_2, \beta_2 \rangle} \beta_2 - \frac{\langle a_4, \beta_3 \rangle}{\langle \beta_3, \beta_3 \rangle} \beta_3$$

$$= t^3 - \frac{\langle t^3, 1 \rangle}{\langle 1, 1 \rangle} \cdot 1 - \frac{\langle t^3, t - \frac{1}{2} \rangle}{\langle t - \frac{1}{2}, t - \frac{1}{2} \rangle} (t - \frac{1}{2}) -$$

$$\frac{\langle t^3, t^2 - t + \frac{1}{6} \rangle}{\langle t^2 - t + \frac{1}{6}, t^2 - t + \frac{1}{6} \rangle} (t^2 - t + \frac{1}{6})$$

$$= t^3 - \frac{\frac{1}{4}}{\frac{1}{1}} \times 1 - \frac{\frac{1}{5} - \frac{1}{8}}{\frac{1}{12}} (t - \frac{1}{2}) -$$

$$\frac{\frac{1}{6} - \frac{1}{5} + \frac{1}{24}}{\frac{1}{180}} (t^2 - t + \frac{1}{6})$$

$$= t^3 - \frac{3}{2}t^2 + \frac{3}{5}t - \frac{1}{20}$$



$$b_1 = \frac{\langle x, \beta_1 \rangle}{\langle \beta_1, \beta_1 \rangle} = \frac{\langle t^5, 1 \rangle}{\langle 1, 1 \rangle} = \int_0^1 t^5 dt = \frac{1}{6}$$

$$b_2 = \frac{\langle x, \beta_2 \rangle}{\langle \beta_2, \beta_2 \rangle} = \frac{\langle t^5, t - \frac{1}{2} \rangle}{\langle t - \frac{1}{2}, t - \frac{1}{2} \rangle} = \frac{\frac{1}{7} - \frac{1}{12}}{\frac{1}{12}} = \frac{5}{7}$$

$$b_3 = \frac{\langle x, \beta_3 \rangle}{\langle \beta_3, \beta_3 \rangle} = \frac{\langle t^5, t^2 - t + \frac{1}{6} \rangle}{\langle t^2 - t + \frac{1}{6}, t^2 - t + \frac{1}{6} \rangle} = \frac{\frac{1}{8} - \frac{1}{7} + \frac{1}{36}}{\frac{1}{180}} = \frac{25}{14}$$

$$\begin{aligned} b_4 &= \frac{\langle x, \beta_4 \rangle}{\langle \beta_4, \beta_4 \rangle} = \frac{\langle t^5, t^3 - \frac{3}{2}t^2 + \frac{3}{5}t - \frac{1}{20} \rangle}{\langle t^3 - \frac{3}{2}t^2 + \frac{3}{5}t - \frac{1}{20}, t^3 - \frac{3}{2}t^2 + \frac{3}{5}t - \frac{1}{20} \rangle} \\ &= \frac{\frac{1}{9} - \frac{3}{16} + \frac{3}{35} - \frac{1}{120}}{\frac{1}{7} - \frac{1}{2} + \frac{69}{100} - \frac{19}{40} + \frac{51}{300} - \frac{3}{100} + \frac{1}{400}} \\ &= 2.7778 \approx \frac{25}{9} \quad (\text{这一步用计算器}) \end{aligned}$$

(3) 见后续程序.

这里

$$\begin{aligned} &b_1 \beta_1(t) + b_2 \beta_2(t) + b_3 \beta_3(t) + b_4 \beta_4(t) \\ &= \frac{1}{6} + \frac{5}{7} \left(t - \frac{1}{2}\right) + \frac{25}{14} \left(t^2 - t + \frac{1}{6}\right) + \frac{25}{9} \left(t^3 - \frac{3}{2}t^2 + \frac{3}{5}t - \frac{1}{20}\right) \\ &= \frac{25}{9} t^3 - \frac{50}{21} t^2 + \frac{25}{42} t - \frac{25}{63} \end{aligned}$$



5. ① 证明 $\left\{ e^{j\frac{2\pi}{N}nk} \right\}_{k=1,2,\dots,(N-1)}$ 是正交基

$$\begin{aligned} & \langle e^{j\frac{2\pi}{N}nk}, e^{j\frac{2\pi}{N}nl} \rangle \\ &= \sum_{n=0}^{N-1} e^{j\frac{2\pi}{N}kn} \overline{e^{j\frac{2\pi}{N}ln}} \\ &= \sum_{n=0}^{N-1} e^{j\frac{2\pi}{N}(k-l)n} \\ &= \begin{cases} \frac{1 - e^{j\frac{2\pi}{N}(k-l)N}}{1 - e^{j\frac{2\pi}{N}(k-l)}} = \frac{1 - e^{j2\pi(k-l)}}{1 - e^{j\frac{2\pi}{N}(k-l)}} = 0, & \text{当 } k \neq l \text{ 时} \\ N, & \text{当 } k = l \text{ 时} \end{cases} \end{aligned}$$

因此: 由 $x[n] = \sum_{k=1}^{N-1} a_k e^{j\frac{2\pi}{N}nk}$

$$= \sum_{l=1}^{N-1} a_l e^{j\frac{2\pi}{N}nl}$$

有: $\langle x[n], e^{j\frac{2\pi}{N}nk} \rangle = \sum_{l=1}^N a_l \langle e^{j\frac{2\pi}{N}nl}, e^{j\frac{2\pi}{N}nk} \rangle$

$$= a_k \cdot N \quad \left(\begin{array}{l} \text{只在 } k=l \text{ 时内} \\ \text{积为 } N \end{array} \right)$$

因此 $a_k = \frac{1}{N} \langle x[n], e^{j\frac{2\pi}{N}nk} \rangle$

$$= \frac{1}{N} \sum_{n=0}^{N-1} x[n] e^{-j\frac{2\pi}{N}nk}$$



6. ① 将 $X_{\cos}(w) = \int_{-\infty}^{+\infty} X(u) \cos(wu) du$ 和

$$X_{\sin}(w) = \int_{-\infty}^{+\infty} X(u) \sin(wu) du \text{ 代入}$$

$$\frac{1}{2\pi} \int_{-\infty}^{+\infty} [X_{\cos}(w) \cos(wt) + X_{\sin}(w) \sin(wt)] dw$$

$$= \frac{1}{2\pi} \int_{-\infty}^{+\infty} dw \int_{-\infty}^{+\infty} X(u) [\cos(wu) \cos(wt) + \sin(wu) \sin(wt)] du$$

~~$$= \frac{1}{2\pi} \int_{-\infty}^{+\infty} du \int_{-\infty}^{+\infty}$$~~

$$= \frac{1}{2\pi} \int_{-\infty}^{+\infty} X(u) du \int_{-\infty}^{+\infty} \cos[w(u-t)] dw$$

$$= \frac{1}{2\pi} \int_{-\infty}^{+\infty} X(u) du \cdot \frac{1}{u-t} \sin[w(u-t)] \Big|_{w=-\infty}^{+\infty}$$

$$= \frac{1}{2\pi} \int_{-\infty}^{+\infty} X(u) du \lim_{w \rightarrow +\infty} \frac{\sin[w(u-t)] - \sin[-w(u-t)]}{u-t}$$

$$= \frac{1}{2\pi} \int_{-\infty}^{+\infty} X(u) du \lim_{w \rightarrow +\infty} \frac{2 \sin[w(u-t)]}{(u-t)}$$

$$= \int_{-\infty}^{+\infty} X(u) du \left[\lim_{w \rightarrow +\infty} \frac{\sin[w(u-t)]}{\pi(u-t)} \right]$$

$$= \int_{-\infty}^{+\infty} X(u) \delta(u-t) du$$

$$= X(t)$$

命题成立



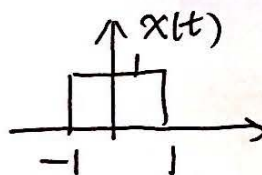
$$(2) X(j\omega) = X \cos(\omega) + j X \sin(\omega)$$

若 $x(t)$ 是实函数, 则有:

$$X \cos(\omega) = \operatorname{Re}\{X(j\omega)\}$$

$$X \sin(\omega) = -\operatorname{Im}\{X(j\omega)\}$$

7. ① 先求一个周期 $x(t)$ 的傅里叶变换



$$\xrightarrow{F} 2 \operatorname{Sa}(\omega) = \frac{2 \sin \omega}{\omega} = X(j\omega)$$

再用公式 $a_k = \frac{1}{T_0} X(jk\omega_0)$, 观察知

$$T_0 = 4, \quad \omega_0 = \frac{2\pi}{T_0} = \frac{\pi}{2}, \quad \text{则}$$

$$a_k = \frac{1}{4} \cdot \frac{2 \sin(\frac{\pi}{2}k)}{k \frac{\pi}{2}} = \frac{\sin(\frac{\pi}{2}k)}{k\pi}$$

因此

$$x(t) = \sum_{k=-\infty}^{+\infty} a_k e^{jk\omega_0 t}$$

$$= \sum_{k=-\infty}^{+\infty} \frac{\sin(k \frac{\pi}{2})}{k\pi} e^{jk \frac{\pi}{2} t}$$

$$= \lim_{k \rightarrow 0} \frac{\sin(k \frac{\pi}{2})}{k \cdot \pi} + \sum_{k=1}^{+\infty} \left[\frac{\sin(k \frac{\pi}{2})}{k\pi} e^{jk \frac{\pi}{2} t} + \frac{\sin(-k \frac{\pi}{2})}{-k\pi} e^{-jk \frac{\pi}{2} t} \right]$$

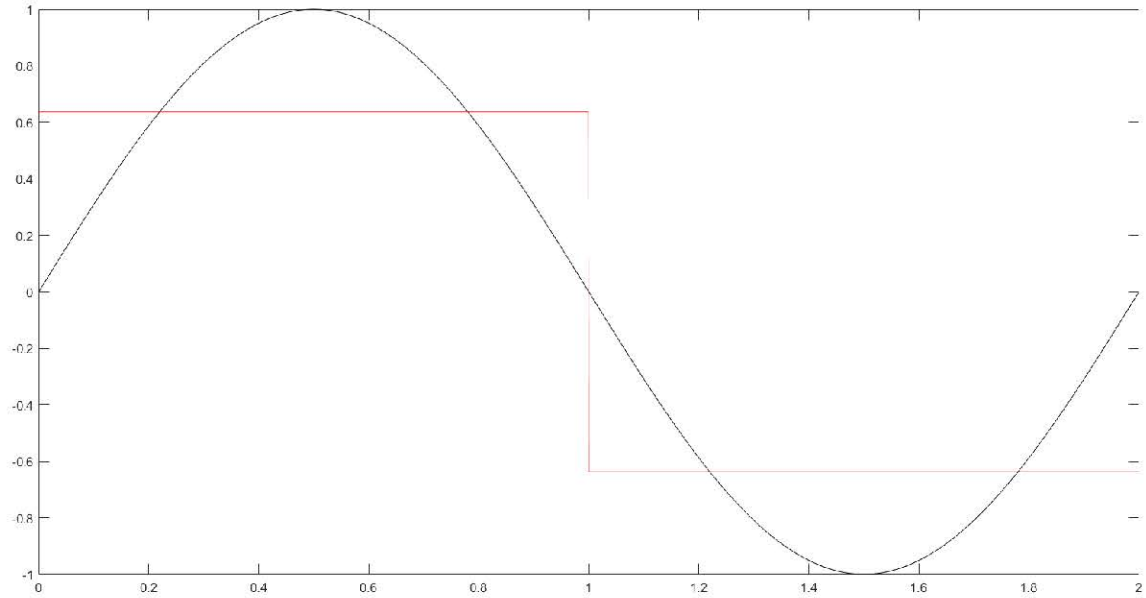
$$= \frac{1}{2} + \sum_{k=1}^{+\infty} \frac{\sin(k \frac{\pi}{2})}{k\pi} (e^{jk \frac{\pi}{2} t} + e^{-jk \frac{\pi}{2} t})$$

$$= \frac{1}{2} + \sum_{k=1}^{+\infty} \frac{2 \sin(k \frac{\pi}{2})}{k\pi} \cos(k \frac{\pi}{2} t)$$



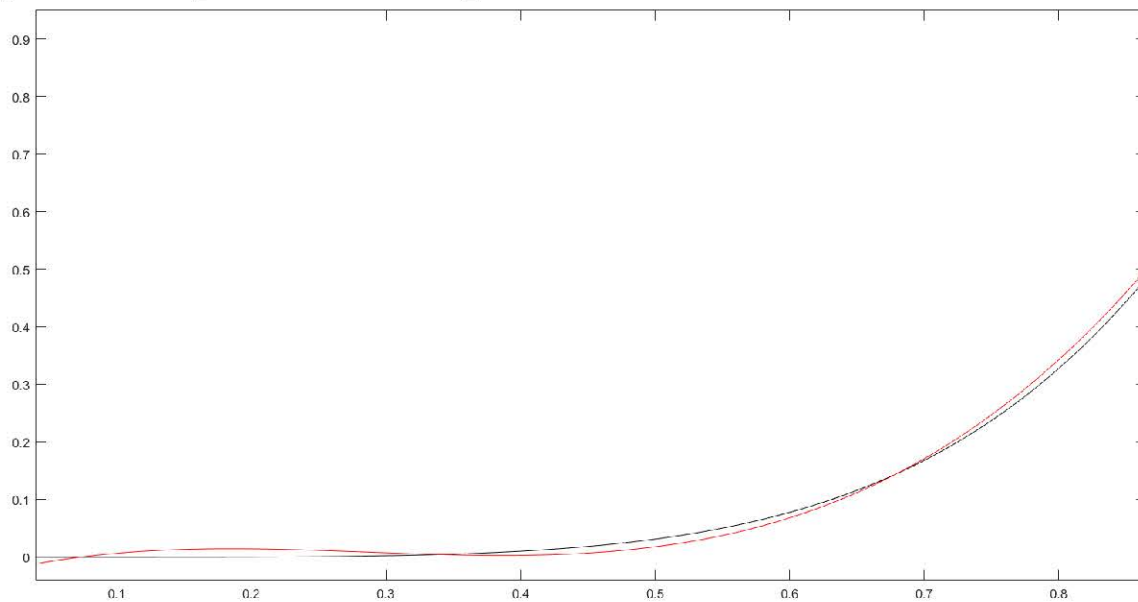
(1) 第2题MATLAB程序

```
t= 0:0.001:2;  
y1 = sin(pi*t);  
y2 = ones(1,length(t))  
y2(length(t)/2:end) = -1;  
y2 = 2*y2/pi;  
plot(t,y1,'k-',t,y2,'r-');
```



(2) 第4题MATLAB程序

```
t = 0:0.0001:1;  
y1 = t.^5;  
y2 = 25*t.^3/9-50*t.^2/21+25*t/42-2/63;  
plot(t,y1,'k-',t,y2,'r-');
```



(3) 第7题MATLAB程序

```
t = -2:0.0001:2;
x = zeros(1,length(t));
for i =1:length(t)
    if t(i)>=-1 && t(i)<=1
        x(i) = 1;
    end;
end;
subplot(2,1,1)
plot(t,x, 'r-');

N =
1000; % Ä±äN=5,10,100,1000 ÷×ÔÔËÐÐÒ»±é;£
xn = 0.5*ones(1,length(t));
for k=1:N
    xn =
xn+2*sin(k*pi/2)*cos(k*pi*t/2)/(k*pi);
end;
subplot(2,1,2);
plot(t,xn, 'r-');
```

